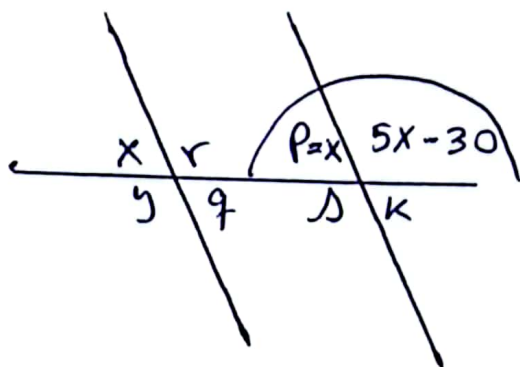


GEOMETRIA

①

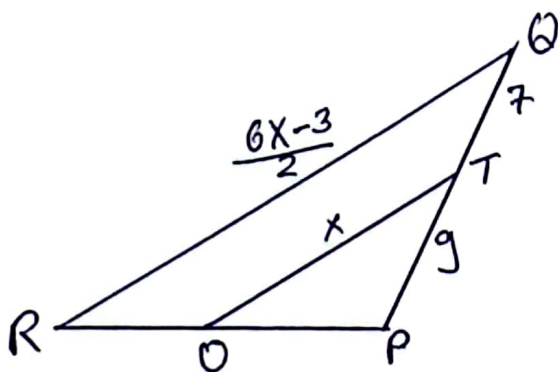


$$\begin{aligned} *) \quad x + (5x - 30) &= 180 \\ 6x &= 210 \\ \underline{x = 35} \end{aligned}$$

$$\begin{aligned} *) \quad x + \Delta &= 180 \\ 35 + \Delta &= 180 \\ \underline{\Delta = 145} \end{aligned}$$

$$\begin{aligned} \text{Luego } 2x + \Delta &= 2(35) + 145 \\ &= 215 \end{aligned}$$

②

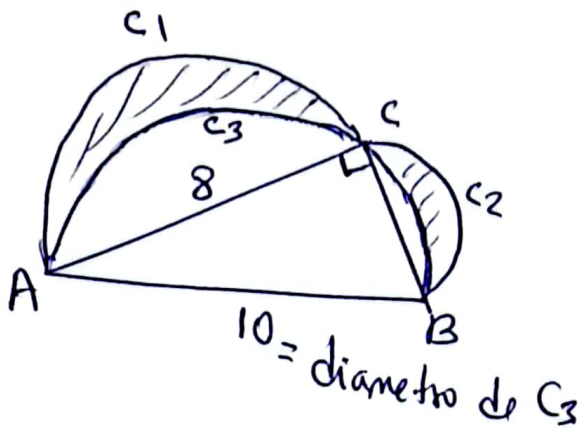


$$\begin{aligned} \frac{RQ}{OT} &= \frac{PT}{TP} \\ \frac{\frac{6x-3}{2}}{x} &= \frac{16}{9} \end{aligned}$$

$$\begin{aligned} 9(6x-3) &= 32x \\ 54x - 27 &= 32x \\ 22x &= 27 \\ x &= \frac{27}{22} \end{aligned}$$

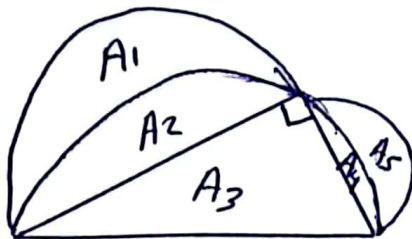
$$\begin{aligned} \text{Luego } 6x &= 6\left(\frac{27}{22}\right) = \frac{81}{11} \end{aligned}$$

③



$$\begin{aligned} \Delta B^2 &= AC^2 + BC^2 \\ 10^2 &= 8^2 + BC^2 \\ \underline{6 = BC} \end{aligned}$$

luego



Para hallar $A_1 + A_5$, se tiene

$$\begin{aligned} \forall) \quad A_1 &= \frac{1}{2} (\pi r^2) - A_2 \\ &= \frac{1}{2} \pi \cdot 4^2 - A_2 \end{aligned}$$

$$\underline{A_1 = \frac{16\pi}{2} - A_2}$$

$$\forall) \quad A_5 = \frac{1}{2} (\pi r^2) - A_4$$

$$A_5 = \frac{1}{2} (\pi \cdot 3^2) - A_4$$

$$\underline{A_5 = \frac{9\pi}{2} - A_4}$$

$$\text{entonces: } A_1 + A_5 = \left(\frac{16\pi}{2} - A_2 \right) + \left(\frac{9\pi}{2} - A_4 \right)$$

$$= \frac{25\pi}{2} - (A_2 + A_4)$$

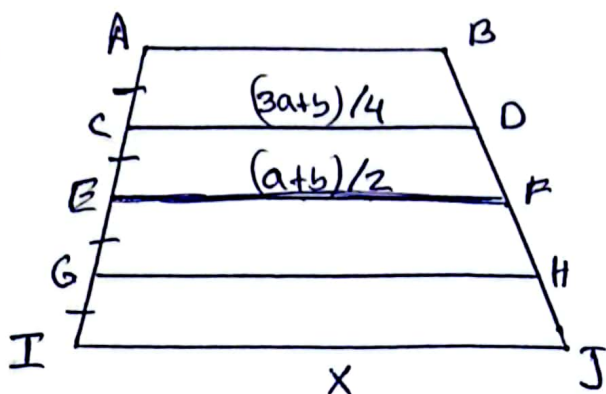
$$= \frac{25\pi}{2} - (A_2 + A_4 + A_3) + A_3$$

$$= \frac{25\pi}{2} - \left(\frac{1}{2} \pi r^2 \right) + \underline{A_3}$$

$$= \frac{25\pi}{2} - \frac{1}{2} \pi \cdot 5^2 + \frac{1}{2} \cdot 6 \cdot 8$$

$$\underline{A_1 + A_5 = 24 \text{ cm}^2}$$

④



De la figura se tiene.

$$\begin{cases} CD = \frac{AB + EF}{2} \\ EF = \frac{AB + IJ}{2} \end{cases}$$

luego

$$\begin{cases} \frac{3a+b}{4} = \frac{AB + \frac{a+b}{2}}{2} \quad \text{--- (1)} \\ \frac{a+b}{2} = \frac{AB + IJ}{2} \quad \text{--- (2)} \end{cases}$$

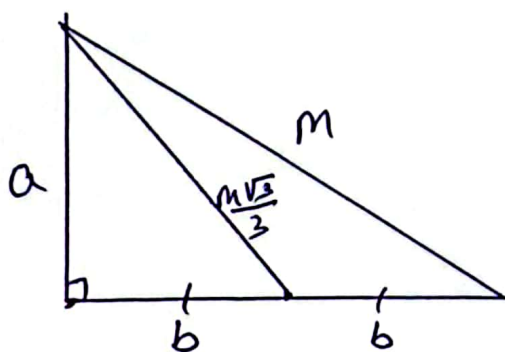
en (1)

$$\frac{3a+b}{2} = AB + \frac{a+b}{2} \Rightarrow \boxed{AB = a}$$

en (2)

$$\begin{aligned} a+b &= AB + IJ \\ a+b &= a + IJ \Rightarrow \boxed{IJ = b} \end{aligned}$$

⑤



$$\begin{cases} a^2 + (2b)^2 = m^2 \\ a^2 + b^2 = \left(\frac{m\sqrt{3}}{3}\right)^2 \end{cases}$$

$$\begin{cases} a^2 + 4b^2 = m^2 \\ a^2 + b^2 = \frac{m^2}{3} \end{cases} \quad \text{--- (1)}$$

$$\begin{aligned} a^2 + 4b^2 &= m^2 \\ -a^2 - b^2 &= -\frac{m^2}{3} \end{aligned}$$

$$3b^2 = \frac{2}{3}m^2 \Rightarrow \boxed{b = \frac{\sqrt{2}}{3}m}$$

luego

$$a^2 + \left(\frac{\sqrt{2}}{3}m\right)^2 = \frac{m^2}{3}$$

$$\boxed{a = \frac{m}{3}}$$

los lados son

$$a = \frac{m}{3}, \quad 2b = \frac{2\sqrt{2}}{3}m$$